

Solar Energy Production in Urban Environments: Integration of Balcony Power Plant into Smart Home Systems

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Abstract— In the context of rising energy costs and increasing demand for electricity in modern urban environments, small-scale residential renewable solutions are gaining importance. This paper presents the implementation of a balcony power plant integrated into the OBLO smart home system, enabling localized photovoltaic energy production and improved energy awareness. The proposed solution supports clean energy generation, energy savings, and real-time monitoring. The system is evaluated using real operational data from an installed device, demonstrating the practical benefits and feasibility of balcony-based solar power generation in contemporary residential settings.

Keywords—Smart home system, Photovoltaic system, Internet of Things, Solar energy

I. INTRODUCTION

Rising electricity prices and increasing energy demand have strengthened the interest in decentralized renewable energy solutions. The global photovoltaic market has experienced significant growth over the past decade due to decreasing technology costs and supportive energy policies [1]. In modern urban environments, where a large portion of the population lives in apartment buildings, conventional rooftop PV installations are often not feasible. As a result, small balcony-based solar units have emerged as a practical option for residential energy production.

Recent studies have analyzed the growing adoption of plug-in and balcony PV systems in urban residential environments, indicating their ability to support distributed renewable energy use in apartment-based housing [2], [3].

Balcony power plants are compact plug-in PV systems designed for individual households. They typically consist of one or two solar panels connected to a microinverter, enabling direct injection of generated electricity into the household electrical network. In this way, households can partially offset their electricity consumption, reduce dependency on the public grid, and increase awareness of their own energy usage patterns. Because they are relatively easy to install and are subject to simplified regulations in many European countries, balcony PV systems have become more common in recent years. Several European studies have examined their performance and energy production, showing that even small installations can cover a noticeable portion of a household's electricity needs [2], [4], [5].

Parallel to these developments, Internet of Things (IoT) technologies and smart home platforms have transformed the way residential energy systems are monitored and managed.

Modern smart home systems provide centralized control, automation, and real-time data visualization of various household devices. Integrating a PV system into a smart home ecosystem enables not only energy production but also intelligent energy management, performance tracking, and improved user interaction. The integration of distributed photovoltaic systems with IoT-based monitoring platforms represents an important research direction within modern residential energy management and smart home systems [6].

The OBLO smart home system represents one such platform that supports communication between controllers, cloud services, and mobile applications. Using its existing communication and data exchange infrastructure, renewable energy monitoring functionalities can be incorporated into the platform. This integration enables users to observe energy production in real time, analyze historical data, and make informed decisions regarding energy consumption.

This paper presents the implementation and integration of a balcony power plant into the OBLO smart home system. The proposed solution combines localized PV energy generation with smart monitoring functionalities, creating a unified residential energy management approach. The system is evaluated using real operational data from an installed PV device, demonstrating the feasibility, efficiency, and practical benefits of such integration in contemporary urban settings.

II. RELATED WORK

The increasing demand for sustainable energy solutions in urban environments has led to significant research in distributed renewable energy systems. In densely populated residential areas, traditional rooftop photovoltaic installations are often limited by space constraints and building ownership structures. As a result, small-scale solutions such as balcony photovoltaic (PV) systems have emerged as an alternative approach for enabling decentralized energy production at the household level.

Several studies have analyzed the performance and feasibility of balcony PV systems. Krauter and Bendfeld [2] evaluated the efficiency and energy yield of photovoltaic microinverters in such systems, while Polański [3] investigated their applicability in large-panel residential buildings. These studies demonstrate that, despite their limited capacity, balcony PV systems can provide meaningful contributions to household energy consumption in urban settings.

In parallel, research on smart home systems has focused on integrating energy monitoring, data acquisition, and optimization capabilities using IoT technologies. Ivanović et al. [7] and Nikolić et al. [8] highlight the importance of scalable data collection and cloud-based processing in smart home environments, enabling efficient handling of large volumes of device-level data. Furthermore, smart home energy management systems support real-time monitoring and optimization of energy usage, improving both system efficiency and user awareness [6].

While these research directions have been extensively explored individually, the integration of small-scale balcony PV systems into existing smart home platforms remains less addressed. There is limited work that combines real-time PV energy production monitoring with smart home infrastructures using real-world operational data.

This paper addresses this gap by presenting the integration of a balcony photovoltaic system into the OBLO smart home platform, enabling real-time monitoring, data aggregation, and analysis of energy production within a unified residential system.

III. SYSTEM ARCHITECTURE

The OBLO system represents a smart home automation platform based on a distributed architecture. It consists of the following core components:

- Controller (Gateway) – the central system unit
- Cloud service
- Client application (Web or mobile)
- Peripheral devices

The Controller represents the central component of the system. Its primary responsibilities include collecting data, managing and controlling peripheral devices, processing and distributing system information. Due to the heterogeneity of connected devices, the controller supports multiple communication protocols, including IP-based communication, BLE, ZigBee, Z-Wave, and Matter, depending on the hardware. The user can communicate with the controller via the local network or through the cloud service. In this implementation, the controller receives production data from the balcony photovoltaic system, such as instantaneous power output and cumulative generated energy. These values are processed and made available to the client application for monitoring and analysis.

The cloud service enables remote access to the controller outside the local network environment. The Client application sends an HTTP/MQTT request to the cloud service, and the cloud forwards commands to the controller. The controller responds, and the cloud relays the response back to the client application. In the case of the integrated balcony PV system, the cloud allows users to monitor energy production regardless of their physical location. In the cloud layer, data from multiple devices and users is processed and stored using a service-based architecture, where each device is associated with a unique identifier. This enables scalable data handling and ensures that energy production and consumption data can be managed independently for each user within the system, supporting multi-user environments and distributed deployments.

The client application serves as the user interface of the system, and it is implemented as a mobile application or a web-based interface. The application can establish communication either directly with the controller via MQTT (local mode) or indirectly through the cloud service using HTTP and MQTT (remote mode). Its main functions within this research include monitoring real-time PV energy production, displaying hourly, daily, or monthly energy statistics, enhancing user awareness of household energy generation.

Peripheral devices in the OBLO system are categorized as sensors (collecting data) or actuators (executing commands). The balcony PV system functions primarily as an energy sensor, measuring electricity production parameters such as instantaneous power output (W), generated energy (kWh).

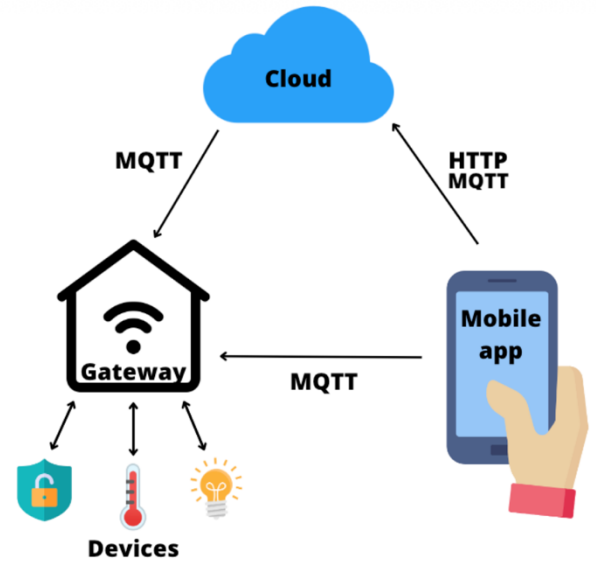


Fig. 1. OBLO smart home system

By integrating the balcony photovoltaic unit as an energy-producing peripheral device, the OBLO platform extends from traditional smart home automation toward distributed residential energy monitoring and management.

IV. INTEGRATION OF THE BALCONY PHOTOVOLTAIC SYSTEM INTO OBLO

The integration of the PV system into the OBLO smart home platform was implemented using a local data acquisition architecture based on the open-source OpenDTU project [9]. The objective of the integration was to abstract the PV installation as a native OBLO energy-producing device while preserving architectural consistency with existing system components.

The system consists of two solar panels connected to a microinverter with a total installed capacity of 600 W. The inverter communicates wirelessly using a proprietary 2.4 GHz RF protocol. To access inverter data, a Data Transfer Unit (DTU) was implemented using the OpenDTU project. The DTU hardware is based on:

- ESP8266 (Wemos D1 mini) microcontroller
- NRF24L01+ 2.4 GHz RF module

The system operates without battery storage, meaning that all generated energy is either consumed in real time or injected into the household electrical network. This configuration is typical for plug-in balcony PV systems, prioritizing simplicity and ease of installation over energy storage capabilities.

The DTU receives operational measurements from the inverter via RF communication and exposes them through an embedded HTTP server. This server provides a REST API interface that allows retrieval of inverter data in JSON format. The DTU device is automatically discovered within the local network using the mDNS protocol. This enables dynamic identification without requiring static IP configuration.

Communication between the OBLO controller and the DTU is performed exclusively over the local network using HTTP REST requests. The implemented setup performs periodic polling at 10-minute intervals.

Each polling cycle consists of:

- Execution of an HTTP GET request to the DTU REST endpoint
- Reception of JSON-formatted inverter data
- Parsing and extraction of relevant electrical parameters

This lightweight polling mechanism introduces negligible network load and ensures near real-time monitoring performance suitable for residential applications.

The internal processing sequence is illustrated in Fig. 2, showing the periodic command execution, HTTP request handling, response parsing, and service population process. This implementation ensures full compatibility with existing OBLO device abstraction mechanisms.

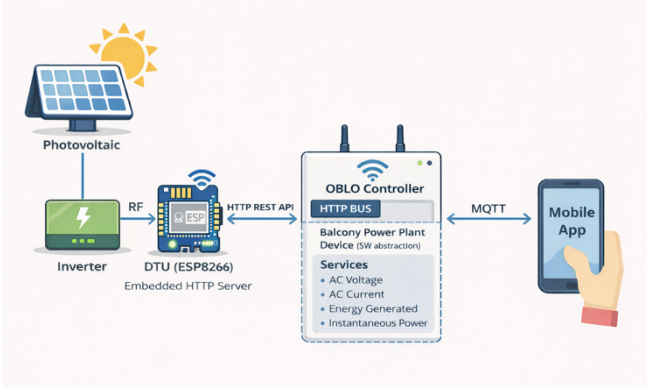


Fig. 2. Communication between PP and OBLO systems

Based on the data provided by the OpenDTU REST API and the functional requirements of the OBLO smart home platform, a subset of relevant electrical parameters was selected for integration and exposure within the system. The selected properties represent the key indicators of photovoltaic system performance and enable meaningful monitoring and analysis. The following parameters were implemented as OBLO device properties:

- AC Voltage (grid voltage)
- AC Current (generated current)

- Total Energy Generated (kWh)
- Instantaneous Power Output (W)

Based on the measurement data delivered to the OBLO controller, a subset of relevant electrical parameters was processed and exposed within the system. The gateway periodically retrieves production values, parses the received responses, and maps the extracted measurements to internal OBLO services and properties.

V. TESTING AND RESULTS

The evaluation was conducted over a six-month period, from September 2025 to February 2026, using real operational data from an installed system. Performance was evaluated using metrics such as total energy production, financial savings, carbon emission reduction, specific yield, and capacity factor. Data were monitored through the OBLO smart home mobile application.

The experimental data were collected from a real balcony photovoltaic installation located in Suhr, Switzerland (47.37°N, 8.08°E). This region is characterized by a temperate Central European climate with significant seasonal variations in solar irradiation.

The average annual global horizontal solar irradiation in Switzerland typically ranges between 1000 and 1400 kWh/m², as reported by national solar resource assessments and PV performance studies [10]. During winter months (December–January), irradiation levels are substantially lower due to reduced daylight duration and lower solar elevation angles. This climatic characteristic directly influences the seasonal performance of small-scale PV systems.

Energy production data were collected automatically via the microinverter monitoring interface and transmitted to the OBLO controller. The controller processed the incoming values and stored them in the system database. Data were sampled at regular time intervals and aggregated into hourly, daily, and monthly statistics within the OBLO cloud platform. This approach ensured continuous monitoring without manual intervention and allowed reliable long-term performance evaluation.

Throughout the monitoring period, the integrated system demonstrated stable and reliable operation.

No significant data loss was recorded during the six-month observation interval. MQTT-based communication between the inverter interface, OBLO controller, cloud service, and mobile application remained stable. Data synchronization operated continuously without interruption.

Measured latency between energy production measurement and visualization in the client application remained negligible for residential monitoring purposes.

The results confirm that the PV unit can be successfully abstracted as a virtual energy sensor within the OBLO platform, extending traditional smart home automation toward distributed residential energy monitoring and management.

The recorded monthly production values are shown in Fig. 3 and summarized in table I.

TABLE I. MONTHLY ENERGY PRODUCTION

Month	Energy Production (kWh)
September	65.0
October	42.0
November	23.9
December	12.2
January	14.4
February	21.1

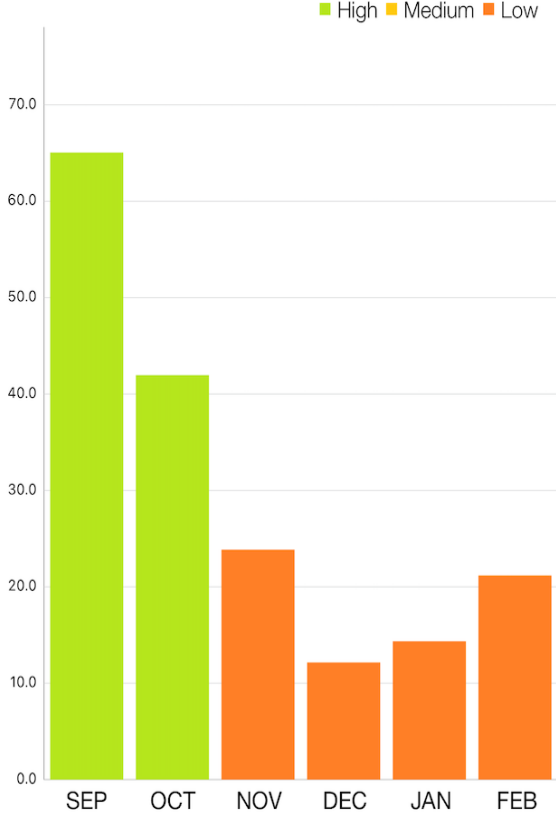


Fig. 3. Monthly energy production

The results clearly demonstrate a strong seasonal dependency of PV energy generation. The highest production was recorded in September, reaching 65 kWh, which corresponds to favorable solar irradiation and longer daylight duration.

From October onward, a gradual decrease in energy production can be observed due to reduced solar intensity and shorter days. The lowest production occurred during December (12.2 kWh), which is expected for winter conditions in Central Europe.

Daily production profiles further confirm the dependence on weather conditions and solar exposure. On high-irradiation days, daily energy production reached values close to 3.5 kWh in September as shown in Fig. 4.

In contrast, during winter months, daily production often remained below 1 kWh, with some days falling below 0.5 kWh due to cloud coverage and low solar elevation angles.

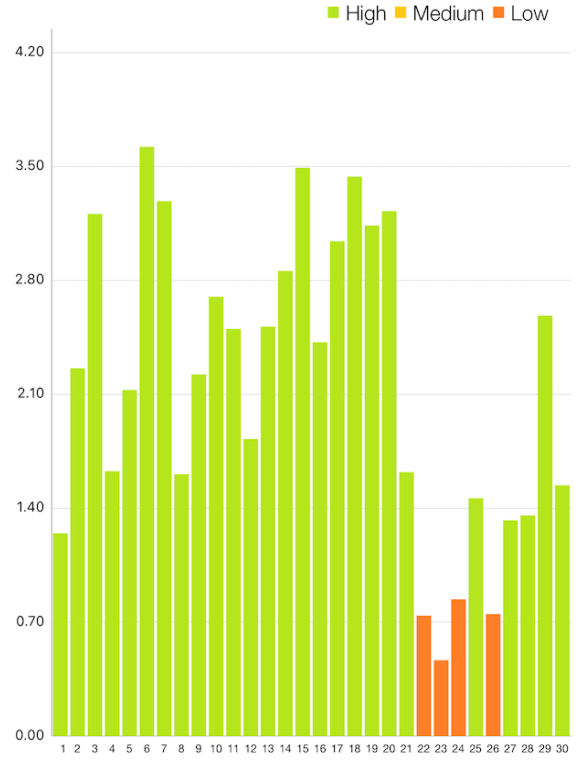


Fig. 4. Daily production in September

Based on the total six-month production of 178.6 kWh and considering the average electricity price in Switzerland (approximately 0.30 CHF/kWh), based on publicly available electricity tariff statistics for Swiss residential consumers [11], the estimated financial savings for this period is shown in Equation (1).

$$178.6 \text{ kWh} \times 0.30 \text{ CHF/kWh} = 53.58 \text{ CHF} \quad (1)$$

Based on the average emission factor of the Swiss electricity mix (approximately 100 g CO₂-eq per kWh), according to national energy mix emission assessments [12], the total six-month energy production of 178.6 kWh results in avoided greenhouse gas emissions, as shown in Equation (2).

$$178.6 \text{ kWh} \times 100 \text{ g CO}_2\text{-eq/kWh} = 17.86 \text{ kg CO}_2\text{-eq} \quad (2)$$

The specific yield represents the amount of electrical energy generated per unit of installed photovoltaic capacity, enabling performance comparison independently of system size. The specific yield of the installation during the observed six-month period is calculated in Equation (3).

$$178.6 \text{ kWh} / 0.6 \text{ kWp} = 298 \text{ kWh/kWp} \quad (3)$$

The theoretical maximum energy production over the six-month observation period (approximately 4380 hours) is calculated in Equation (4), assuming continuous operation at the rated power of 0.6 kW. Based on this value, the capacity factor of the system is determined in Equation (5) as the ratio between the actual measured energy production and the theoretical maximum output. The obtained capacity factor of 6.8% reflects realistic operating conditions, including day-night cycles, weather variability, and installation constraints typical for balcony PV systems.

$$0.6 \text{ kW} \times 4380 \text{ h} = 2628 \text{ kWh} \quad (4)$$

$$178.6 \text{ kWh} / 2628 \text{ kWh} \times 100 \% = 6.8 \% \quad (5)$$

The calculated specific yield and capacity factor reflect the seasonal observation period, which mainly covers autumn and winter months characterized by lower solar irradiation. Consequently, the obtained capacity factor of 6.8% should be interpreted as a seasonal performance indicator rather than a representative annual value.

Despite installation constraints typical for balcony PV systems, such as limited installation area, non-optimal panel orientation, and potential shading from surrounding buildings, the measured production values demonstrate that small-scale plug-in photovoltaic systems can still provide meaningful contributions to household electricity consumption in urban residential environments. The obtained specific yield of 298 kWh/kWp is within the expected range for such systems reported in the literature [2], [3].

VI. DISCUSSION

The obtained results confirm that balcony photovoltaic systems can provide measurable energy, economic, and environmental benefits in urban residential environments. The specific yield of 298 kWh/kWp and the capacity factor of 6.8% reflect realistic operating conditions during autumn and winter periods.

Compared to traditional rooftop PV installations, balcony systems have lower overall energy output due to limited installation area and non-optimal orientation. However, their accessibility and ease of installation make them a practical solution for apartment-based housing.

The integration with the OBLO smart home platform enables continuous monitoring and improved user awareness, representing an added value compared to standalone PV systems. Furthermore, such integration creates opportunities for future energy optimization based on production data.

The system performance is influenced by typical urban constraints, including weather variability, shading effects, and limited installation space. Additionally, the evaluation is based on a single installation and a six-month observation period, which may not fully represent long-term annual performance.

VII. CONCLUSION

In the context of rising energy costs and growing environmental awareness, decentralized renewable energy solutions are becoming increasingly relevant in urban environments. The integration of a balcony PV system into the OBLO smart home platform shows that small-scale solar installations can function not only as standalone energy sources, but also as integrated elements of a smart residential system.

The presented results confirm that even low-capacity balcony power plants can generate measurable economic and environmental benefits. With 178.6 kWh of energy produced over a six-month period, the system contributed to both financial savings and reduced carbon emissions. Beyond energy generation itself, the integration within the OBLO

platform enabled continuous monitoring, data analysis, and improved user awareness of household energy production.

The study shows that combining IoT-based smart home infrastructures with PV systems represents a practical and scalable approach for modern apartment buildings, where traditional rooftop installations are often not practical. Rather than serving as an additional renewable source, the integrated balcony power plant becomes an active element of distributed residential energy management.

Future work could focus on extending the monitoring period to enable full annual and multi-year performance evaluation. Additional research may explore energy production estimation based on historical generation data and local weather conditions. Furthermore, deeper integration with smart home automation logic could allow better coordination between energy production and household consumption, so that certain devices can operate when solar generation is higher.

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